

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JNANA SANGAMA",MACHHE, BELAGAVI-590018



**Project Report
On**

Design of an IMU-Locked Multimodal Edge Device for Structural Health Prognostics using Vision-Acoustic Sensor Fusion

Submitted in partial fulfillment of the requirements for the VI semester

Bachelor of Engineering

in

Artificial Intelligence and Machine Learning

of

Visvesvaraya Technological University, Belagavi.

by

Basukiran D (1CD23AI019)

Durga Prasad A (1CD23AI027)

Harshith H S (1CD23AI039)

Dhanush Gowda(1CD23AI038)

Under the Guidance of

Prof. Sudhanshu Rakesh Katarey

Assistant Professor Dept. of AIML



Department of Artificial Intelligence and Machine Learning

CAMBRIDGE INSTITUTE OF TECHNOLOGY

AN AUTONOMOUS INSTITUTION,

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CAMBRIDGE INSTITUTE OF TECHNOLOGY

AN AUTONOMOUS INSTITUTION

K.R.Puram, Bangalore-560 036

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



CERTIFICATE

Certified that **Mr. Basukiran D , Mr. Durga Prasad A , Mrs. Harshith H S and Mrs. H D Dhanush Gowda** bearing **USN: 1CD23AI019, 1CD23AI027, 1CD23AI039 and 1CD23AI038** respectively, are bonafide students of **Cambridge Institute of Technology**, have successfully completed **Phase I** of the project entitled “**Design of IMU-Locked Multimodal Edge Device for Structural Health Prognostics using Vision-Acoustic Sensor Fusion**” in partial fulfillment of the requirements for VI semester **Bachelor of Engineering in Artificial Intelligence and Machine Learning** of **Visvesvaraya Technological University, Belagavi** during academic year 2025-2026. It is certified that all Corrections/Suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

Project Guide
Mr.Sudhanshu Rakesh Katarey
Dept. of AIML, CITech

Project Co-ordinator
Mr Sudhanshu Rakesh Katarey
Dept. of AIML, CITech

Head of the Dept.
Dr. Varalatchoumy. M
Dept. of AIML, CITech

Principal
Dr. B. V. Ravishankar
CITech

Signature with date

DECLARATION

We, **Mr. Basukiran D , Mr. Durga Prasad A , Mrs. Harshith H S and Mrs. H D Dhanush Gowda** bearing **USN: 1CD23AI019, 1CD23AI027, 1CD23AI039 and 1CD23AI038** respectively, are the students of VI semester, Artificial Intelligence and Machine Learning, Cambridge Institute of Technology, hereby declare that Phase I of the project entitled “**Design of IMU-Locked Multimodal Edge Device for Structural Health Prognostics using Vision-Acoustic Sensor Fusion**” has been carried out by us and submitted in partial fulfillment of the course requirements of VI semester **Bachelor of Engineering in Artificial Intelligence and Machine Learning** as prescribed by **Visvesvaraya Technological University, Belagavi**, during the academic year 2025-2026.

We also declare that, to the best of our knowledge and belief, the work reported here does not form part of any other report on the basis of which a degree or award was conferred on an earlier occasion on this by any other student.

Date:	Name	USN	Signature
Place: Bangalore	Basukiran D	1CD23AI019	
	Durga Prasad A	1CD23AI027	
	Harshith H S	1CD23AI039	
	H D Dhanush Gowda	1CD23AI038	

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Basukiran D(1CD23AI019)

Durga Prasad A(1CD23AI027)

Harshith H S(1CD23AI039)

H D Dhanush Gowda(1CD23AI038)

ABSTRACT

This project introduces a fully offline, handheld multimodal Structural Health Monitoring (SHM) system for real-time, on-site infrastructure assessment in resource-limited settings. Built on a Raspberry Pi 5 edge device, it eliminates cloud reliance through a memory-optimized sequential pipeline. The vision branch employs SegFormer-B0 for crack segmentation and topology analysis, while the acoustic branch uses FFT-based signal processing to detect subsurface defects. An IMU-based orthographic lock ensures reliable data acquisition before deterministic sensor fusion integrates both modalities into a unified health score. A temporal prognostic module tracks degradation trends, transforming the system into a predictive maintenance tool for rural deployment.

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CHAPTER 1

INTRODUCTION

Structural Health Monitoring (SHM) has emerged as a critical field in ensuring the safety and longevity of civil infrastructure. Bridges, buildings, and public facilities in many developing regions are rapidly aging, yet conventional inspection methods remain limited. Manual inspections are subjective, labor-intensive, and prone to human error, while modern AI-based systems often rely solely on visual data, leaving subsurface defects undetected. Furthermore, most advanced diagnostic solutions depend on cloud connectivity, making them unsuitable for rural or remote environments where internet access is unreliable.

This project addresses these challenges by designing a fully offline, handheld multimodal edge device that integrates computer vision and acoustic sensing. The system leverages a Raspberry Pi 5 to run a memory-optimized pipeline, combining SegFormer-B0-based crack segmentation with FFT-driven acoustic analysis for subsurface defect detection. To ensure accuracy, an IMU-based orthographic lock enforces strict device alignment, eliminating perspective distortion. The outputs from both modalities are fused into a unified structural health score, while a temporal prognostic module tracks degradation trends over time.

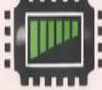
By merging diagnostic precision with predictive capabilities, this architecture transforms infrastructure monitoring from reactive inspection into proactive maintenance. Its low-cost, scalable, and cloud-independent design makes it particularly suited for deployment in resource-constrained environments, offering a practical solution to safeguard critical infrastructure in rural and remote regions.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Literature

Table 2.1 Literature review

Title & Year	Approach	Key Results	Key Results	Gap
YOLOv5 UAV Crack Detection (2021)	 Object Detection (YOLOv5)	Fast crack detection from drone images	No geometry (length/width)	No geometry (length/width) 
U-Net Hairline Fracture (2022)	 Semantic Segmentation (U-Net)	Pixel-perfect crack maps	Heavy, needs GPU/cloud	
SegFormer Civil Monitoring (2023)	 Vision Transformer (SegFormer-B0)	Better topology detection	Assumes perfect angles	
Stereo Vision Perspective Fix (2020)	 Stereo Depth	Corrected image distortion	Expensive hardware	
Acoustic Tap Testing (2019)	 Acoustic NDT (FFT)	Detects internal defects	No surface analysis	
Edge CNN (MobileNet) SHM (2022)	 Edge AI (Raspberry Pi)	Offline crack classification	Only binary output	0/1 
Multimodal Fusion (2023)	 Camera + Ultrasonic	High diagnostic accuracy	High memory usage	
Crack Growth Tracking (2021)	 Temporal Modeling	Tracks crack growth (Δ)	Fixed installation	
IoT Memory Optimization (2022)	 Sequential Execution	Avoids memory crashes	Not for heavy SHM data	

1. YOLOv5 UAV Crack Detection (2021)

- Method: Object detection using YOLOv5 on drone images.
- Strength: Very fast at spotting cracks.
- Limitation: Can't measure crack geometry (length/width).

2. U-Net Hairline Fracture (2022)

- Method: Semantic segmentation with U-Net.
- Strength: Produces pixel-perfect crack maps.
- Limitation: Computationally heavy, requires GPU/cloud.

3. SegFormer Civil Monitoring (2023)

- Method: Vision Transformer (SegFormer-B0).
- Strength: Better at detecting crack topology.
- Limitation: Assumes ideal camera angles, less robust in real-world.

4. Stereo Vision Perspective Fix (2020)

- Method: Stereo depth imaging.
- Strength: Corrects distortion from perspective.
- Limitation: Needs expensive hardware setups.

5. Acoustic Tap Testing (2019)

- Method: Acoustic non-destructive testing (FFT).
- Strength: Detects internal defects invisible to cameras.
- Limitation: No surface crack analysis.

6. Edge CNN (MobileNet) SHM (2022)

- Method: Lightweight CNN on Raspberry Pi.
- Strength: Works offline, portable.
- Limitation: Only binary output (crack/no crack).

7. Multimodal Fusion (2023)

- Method: Combines camera + ultrasonic data.
- Strength: Very high diagnostic accuracy.
- Limitation: High memory usage, not edge-friendly.

8. Crack Growth Tracking (2021)

- Method: Temporal modeling over time.
- Strength: Tracks crack growth (Δ).
- Limitation: Requires fixed installation, not mobile.

9. IoT Memory Optimization (2022)

- Method: Sequential execution for IoT devices.
- Strength: Prevents memory crashes.
- Limitation: Can't handle heavy SHM datasets.

2.2 RESEARCH GAP IDENTIFICATION

Table 2.2 Research Gap Identification table

Research Gap Area	Current State of the Art (Limitation)	How This Project Solves It
1. The Single-Modality Blindspot Existing AI systems rely exclusively on cameras (vision). They are completely blind to internal structural integrity and will classify a visually painted but hollow wall as "safe" (False Negative).	Existing AI systems rely exclusively on cameras (vision). They are completely blind to internal structural integrity and will classify a visually painted but hollow wall as "safe" (False Negative). 	Intruduces a synchronized Multimodal Fusion Engine; pairing the vision camera with a MEMS microphone (FFT Acoustic analysis) to assess both surface topology and subsurface voids simultaneously.
2. Uncontrolled Acquisition (Perspective Distortion) Handheld diagnostic tools assume the operator holds the camera perfectly flat. Uncontrolled angles mathematically compress and distort the physical crack geometry, leading to inaccurate severity scores.	Handheld diagnostic tools assume the operator holds the camera perfectly flat. Uncontrolled angles mathematically compress and distort the physical crack geometry, leading to inaccurate severity scores. 	Implements a strichardware-level IMU Orthographic Lock. The system reads pitch & roll via I2C and physically blocks data acquisition if the device tilt exceeds 5°.
3. The Bounding-Box Fallacy Standard edge-AI deployments rely on object detection models (like YOLO) that merely draw crude boxes around cracks, failing to extract the specific geometry	Replaces object-detection with Topological Semantic Segmentation (SegFormer-B0). Extracting mathematically precise geometry (length, width, branching) to calculate a Topology Severity Index. 	Deploys the entire architecture 100% offline on a Raspberry Pi 5. Utilizes a Sequential Pipeline to prevent RAM overflow, allowing complex AI and DSP to run on constrained edge hardware.
4. Cloud Dependency & Resource Limitations Advanced Transformer AI models are computationally heavy, requiring persistent 5G/W-Fi	Current SHM devices treat every scan as an isolated event ("snapshot" diagnostics), failing to track the historical progression of the structural degradation over time. 	Integrates an onboard Spatial-Temporal Database. By cross-referencing local JSON files, the device calculates the exact Delta (Δ) growth rate, enabling true prognostic predictions.

1. The Single-Modality Blindspot

- Problem: Current AI systems rely only on cameras. They miss internal defects — for example, a hollow wall painted to look fine would be wrongly classified as “safe.”
- Solution: This project fuses vision + acoustic sensing. A camera checks surface cracks, while a MEMS microphone with FFT analysis detects subsurface voids. Together, they cover both external and internal integrity.

2. Uncontrolled Acquisition (Perspective Distortion)

- Problem: Handheld tools assume the operator holds the camera perfectly flat. Tilted angles distort crack geometry, leading to wrong severity scores.
- Solution: A hardware-level IMU Orthographic Lock. It reads tilt (pitch & roll) and blocks data capture if the device is angled more than 5°. This ensures geometric accuracy.

3. The Bounding-Box Fallacy

- Problem: Object detection models like YOLO only draw boxes around cracks. They don’t extract precise geometry (length, width, branching).
- Solution: Uses SegFormer-B0 semantic segmentation to get exact crack topology. It calculates a Topology Severity Index. Runs fully offline on Raspberry Pi 5 with a sequential pipeline to avoid RAM overflow.
-

4. Cloud Dependency & Resource Limitations

- Problem: Transformer models are heavy, often requiring cloud/5G. Current SHM devices only give “snapshot” diagnostics, not long-term progression.
- Solution: Adds an onboard spatial-temporal database. Local JSON files track crack growth (Δ) over time, enabling prognostic predictions instead of just one-time checks.

2.3 Motivation

The motivation behind your project lies in its potential to transform how structural health monitoring is carried out in real-world environments. Aging infrastructure poses serious risks, especially in rural and resource-constrained areas where advanced diagnostic tools are unavailable. Traditional methods are either subjective, limited to surface-level inspection, or dependent on cloud connectivity, making them impractical in such settings. By designing an offline, multimodal edge device that fuses vision-based crack detection with acoustic subsurface analysis, and ensuring accuracy through an IMU orthographic lock, your system directly addresses these gaps. The addition of prognostic tracking elevates the tool from a simple diagnostic scanner to a predictive maintenance solution, enabling communities to anticipate and prevent failures rather than react to them. Ultimately, the motivation is to create a low-cost, scalable, and reliable solution that empowers safer infrastructure management, protects lives, and extends the lifespan of critical structures in places where it is needed most.

2.4 Problem Statement

The growing concern over the safety of aging infrastructure highlights the inadequacy of current diagnostic approaches, particularly in rural and resource-constrained environments. Manual inspections, though widely practiced, are inherently subjective, time-consuming, and prone to human error, often leading to inconsistent evaluations. Automated AI systems that rely solely on visual data are limited to surface-level detection, leaving subsurface defects such as delamination and hollowness undetected. Furthermore, many advanced diagnostic tools depend on cloud-based processing, which is impractical in areas with poor or no internet connectivity. Handheld devices without alignment control introduce perspective distortion, resulting in inaccurate crack geometry and unreliable severity scores. Compounding these issues, existing systems lack temporal prognostics, treating each inspection as an isolated snapshot rather than monitoring the progression of damage over time. These limitations create a critical gap in structural health monitoring, underscoring the need for a **robust, offline, multimodal edge device** that integrates vision-based crack segmentation, acoustic subsurface analysis, and IMU-based alignment to deliver accurate, explainable, and predictive assessments. Such a solution would not only enhance diagnostic reliability but also empower communities to proactively manage infrastructure safety and longevity.

2.5 Objectives

1 Hardware Orthographic Lock

Ensure the device is perfectly aligned with the wall to eliminate perspective distortion.

- Integrate an **IMU sensor** for orientation
- Lock orthographic view before capture
- Prevent skewed or angled imaging

2 Topological Vision Extraction

Capture precise crack geometry using advanced segmentation models.

- Deploy **SegFormer** on edge device
- Extract **pixel-level crack topology**
- Avoid reliance on bounding boxes

3 Acoustic Subsurface Assessment

Analyze tap-test responses to detect hidden hollowness or delamination.

- Use **FFT-based signal processing**
- Identify resonance shifts indicating voids
- Map subsurface anomalies

4 Multimodal Sensor Fusion

Combine vision and acoustic scores into a unified risk classification.

- Apply deterministic fusion algorithm
- Generate **explainable risk score (0–10)**
- Ensure transparency in decision logic

5 Prognostic Tracking

Monitor damage progression over time for predictive maintenance.

- Maintain onboard **temporal database**
- Track crack growth rate
- Forecast structural deterioration

6 Offline Operation

Enable full functionality without cloud dependency.

- Run models entirely on-device
- Support rural and remote environments
- Avoid reliance on external servers

7 Low-Cost Scalability

Design for affordability and portability to enable widespread deployment.

- Use **commodity sensors** where possible
- Optimize for **mass production**
- Ensure lightweight, portable form factor

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Architectural/Design Diagram

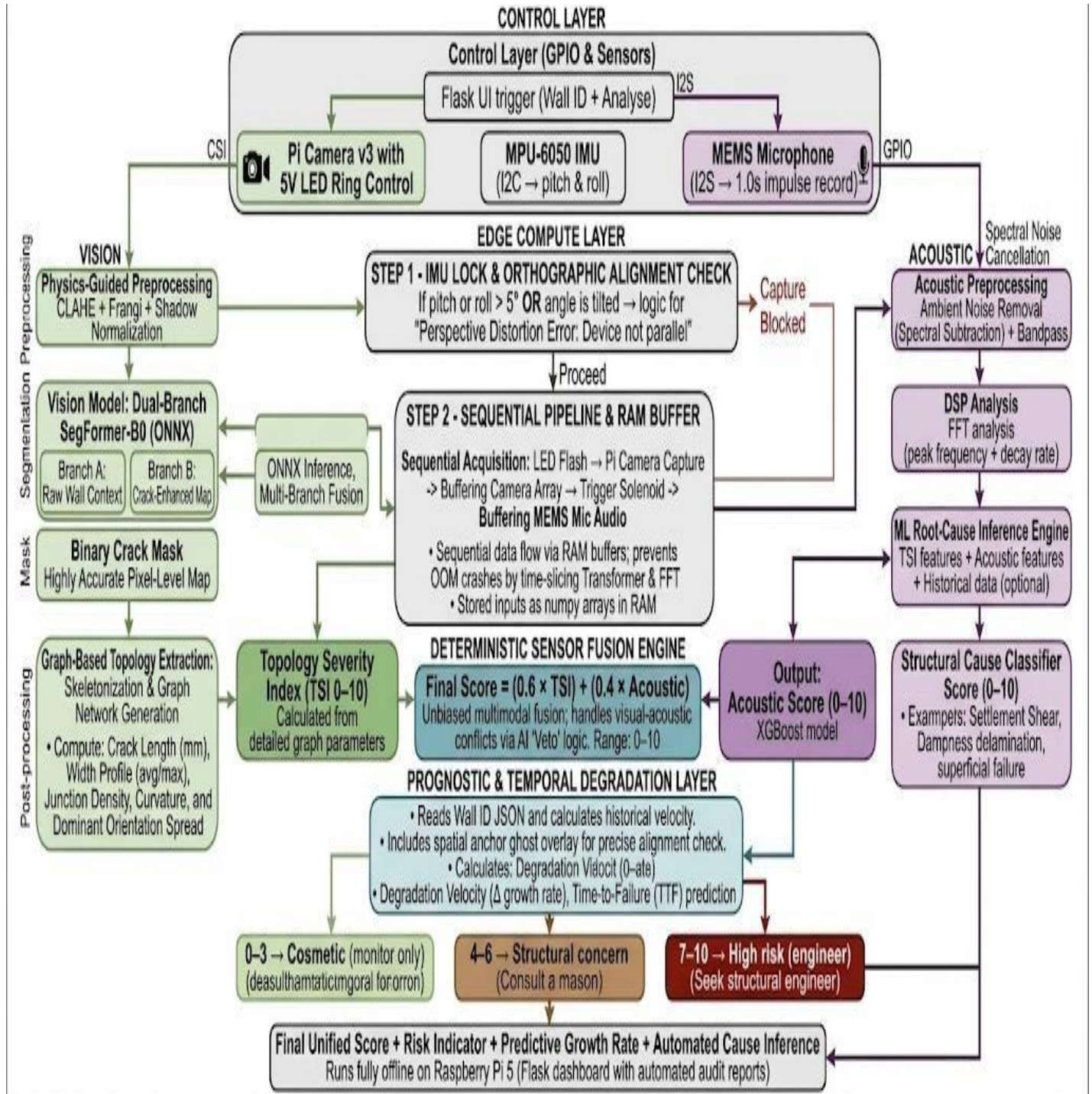


Figure 3.1 Architectural/Design Diagram

Device Alignment (IMU Lock)

- The process begins with the IMU ensuring the device is held orthogonally to the wall.
- This step eliminates perspective distortion and guarantees reliable data capture.

Vision Branch (SegFormer Processing)

- The Pi Camera captures an image of the structure.
- The SegFormer model performs pixel-level crack segmentation, extracting precise crack geometry and topology.
- Output: A vision-based severity score based on crack length, width, and branching.

Acoustic Branch (FFT Analysis)

- The operator performs a tap test on the surface.
- The microphone records the acoustic impulse response.
- FFT-based signal processing detects subsurface anomalies like hollowness or delamination.
- Output: An acoustic resonance score indicating internal structural integrity.

Sensor Fusion (Unified Risk Score)

- Both vision and acoustic scores are combined using a deterministic fusion algorithm.
- The result is a single, explainable risk classification score (0–10).

Prognostic Tracking (Temporal Database)

- Each inspection result is stored in the onboard database with geo-tags and timestamps.
- The system tracks degradation velocity over time, enabling predictive maintenance rather than reactive repairs.

Final Output (Structural Health Report)

- The device generates a unified structural health score and degradation trend.
- Results are displayed locally on the edge device, fully offline, without cloud dependency.

3.2 . Hardware and Software Requirements

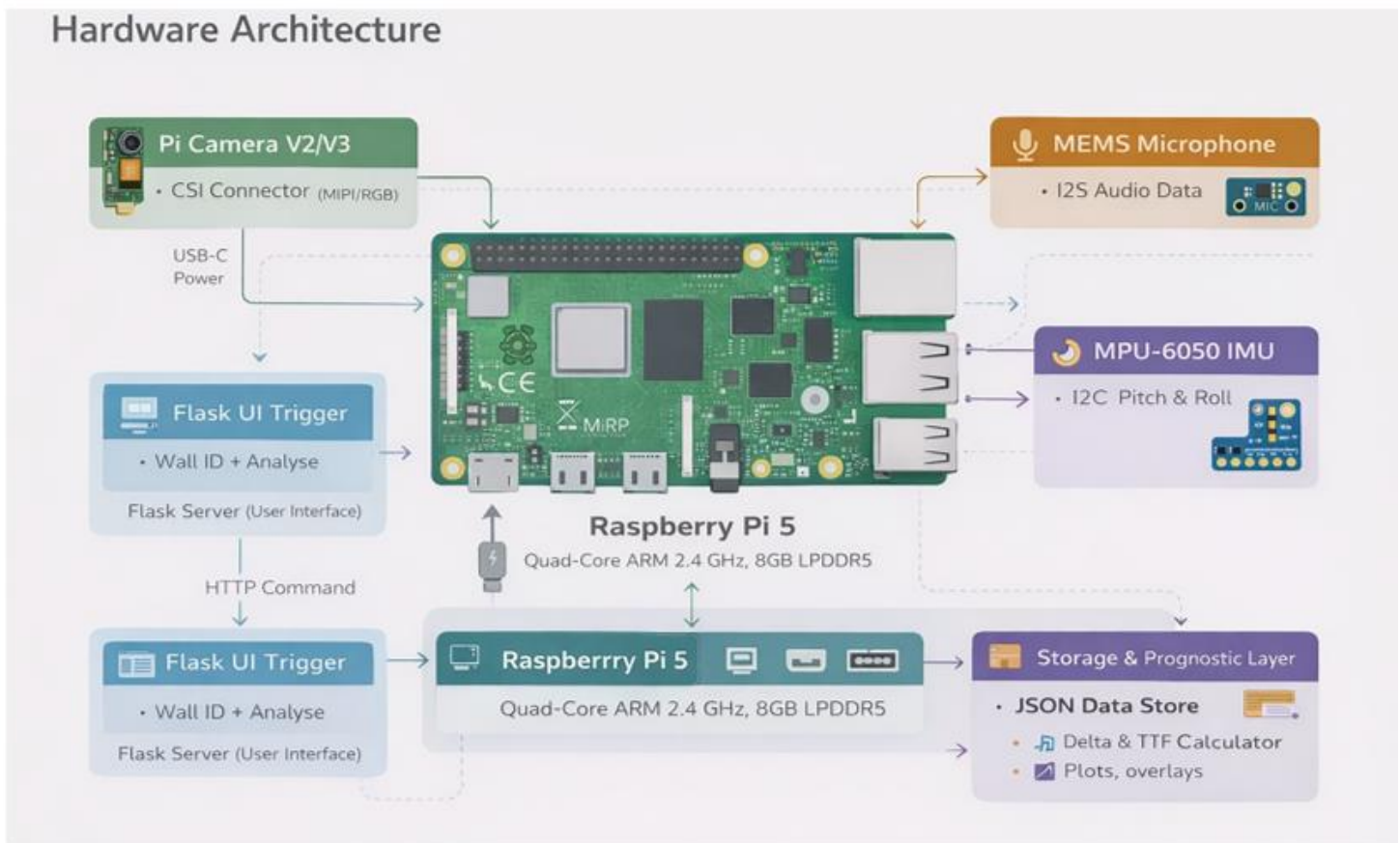


Figure 3.2 Hardware Architecture

Core Components

- Raspberry Pi 5 (Quad-Core ARM 2.4 GHz, 8GB LPDDR5) → The central processing unit, running AI models, acoustic analysis, and the database.
- Pi Camera V2/V3 (CSI connector) → Captures surface crack images. Provides input for segmentation and topology analysis.
- MEMS Microphone (I2S input) → Records acoustic signals for FFT analysis, detecting subsurface voids.
- MPU-6050 IMU (I2C) → Measures pitch & roll. Ensures orthographic lock by blocking acquisition if tilt > 5°.

Software & Data Flow

- Flask UI Trigger (HTTP commands) → User interface layer. Operators can trigger “Wall ID + Analyse” from a web-based dashboard.
- Storage & Prognostic Layer

- JSON Data Store → Saves scan results locally.
- Delta & TTF Calculator → Computes crack growth rate (Δ) and Time-To-Failure predictions.
- Plots/Overlays → Visualizes progression over time.

Integration Logic

1. Sensors → Raspberry Pi

- Camera, microphone, and IMU feed raw data.

2. Processing → AI + DSP

- SegFormer segmentation for geometry.
- FFT acoustic analysis for subsurface defects.
- IMU ensures valid acquisition angles.

3. Storage → Prognostics

- Results stored in JSON.
- Growth tracked over time.

4. UI → Operator

- Flask server provides a simple interface for triggering scans and viewing results.

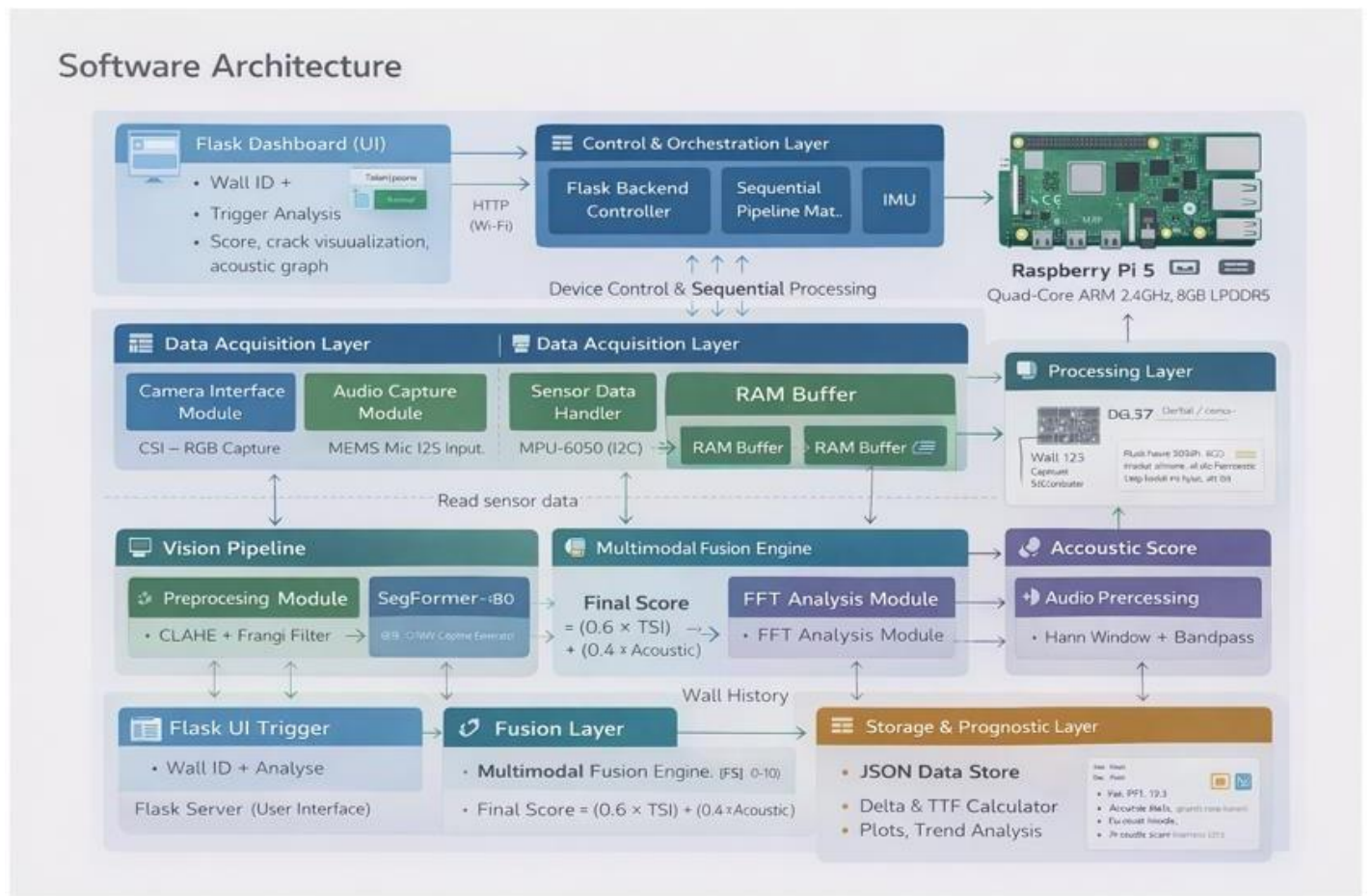


Figure 3.3 Software Architecture

1. Flask Dashboard (UI)

- Operator-facing interface.
- Lets the user trigger analysis, assign wall IDs, and view crack visualizations + acoustic graphs.
- Communicates via HTTP over Wi-Fi.

2. Control & Orchestration Layer

- Flask backend controller manages workflow.
- Sequential pipeline ensures memory stability.
- IMU integration enforces orthographic lock (blocks tilted acquisitions).

3. Data Acquisition Layer

- Camera Interface (CSI – RGB) → captures crack images.
- Audio Capture (MEMS Mic I2S) → records acoustic signals.
- Sensor Handler (MPU-6050 via I2C) → tilt/pitch/roll data.
- RAM Buffers → temporary storage for incoming streams.

4. Processing Layer

- Handles raw data formatting and metadata (wall ID, crack size, sensor status).
- Prepares inputs for vision and acoustic pipelines.

5. Vision Pipeline

- Preprocessing: CLAHE + Frangi filter enhance crack visibility.
- SegFormer-i80: Transformer-based segmentation extracts topology (length, width, branching).

6. Acoustic Score

- Preprocessing with Hann window + bandpass filter.
- FFT analysis detects subsurface anomalies.

7. Multimodal Fusion Engine

- Combines vision + acoustic results.
- Final score formula: $0.6 \times \text{Topology Severity Index (TSI)} + 0.4 \times \text{Acoustic}$.
- Produces a fused diagnostic score (0–10).

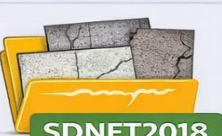
8. Storage & Prognostic Layer

- Local JSON data store.
- Delta (Δ) growth + Time-To-Failure (TTF) calculator.
- Trend analysis with plots and overlays for progression tracking.

3.3 Datasets and Models

Datasets

Table 3.3 Datasets

Dataset Name	Source	Size	Usage in Pipeline
 CRACK500	 Open-Source Academic Benchmark	~ 500 High-Resolution Images with pixel-perfect binary masks	 Vision Branch: Used as core training data for SegFormer-B0 to learn macro-crack topology and precise semantic segmentation.
 DeepCrack	 Open-Source Academic Benchmark	537 Images with multi-scale annotations	 Vision Branch: Supplemental training to improve the model's ability to detect and extract fine, hairline fractures.
 SDNET2018	 Utah State University (Public Dataset)	56,000+ Images of cracked and intact surfaces	 Vision Branch: Used for domain adaptation and validation to ensure accuracy on vertical walls and various masonry textures.
 Custom Acoustic Impulse Responses	 Proprietary (Primary Data collected via project hardware)	1,000+ Audio Arrays (1.0s recordings per tap)	 Acoustic Branch: Used for calibrating the digital Bandpass filter and defining FFT thresholds for “Solid,” “Suspect,” and “Hollow” classifications.

This system is designed to provide a low-cost, portable, and offline solution for infrastructure monitoring, combining vision and acoustic sensing into a unified framework.

Hardware Orthographic Lock An onboard IMU ensures the device is perfectly aligned with vertical walls, eliminating perspective distortion and guaranteeing accurate geometric capture.

Topological Vision Extraction SegFormer runs directly on the edge device, enabling pixel-level crack segmentation. Unlike bounding-box detection, this approach captures the precise topology of cracks, including hairline fractures and complex geometries.

Acoustic Subsurface Assessment FFT-based signal processing analyzes tap-test recordings to classify subsurface conditions as *Solid*, *Suspect*, or *Hollow*. This allows detection of hidden delamination or hollowness that visual inspection alone cannot reveal.

Multimodal Sensor Fusion Vision and acoustic scores are combined through a deterministic algorithm to produce an explainable risk classification on a 0–10 scale. This fusion ensures transparency and reliability in decision-making.

Prognostic Tracking An onboard temporal database records crack growth and acoustic changes over time. By monitoring progression rates, the system supports predictive maintenance and long-term structural health forecasting.

Offline Operation All processing is performed locally on-device, eliminating reliance on cloud services. This makes the system suitable for deployment in rural or remote environments where connectivity is limited.

Low-Cost Scalability The design emphasizes affordability and portability, enabling widespread adoption across diverse infrastructure projects. Its modular architecture ensures scalability from small community buildings to large-scale civil structures.

3.3 Models

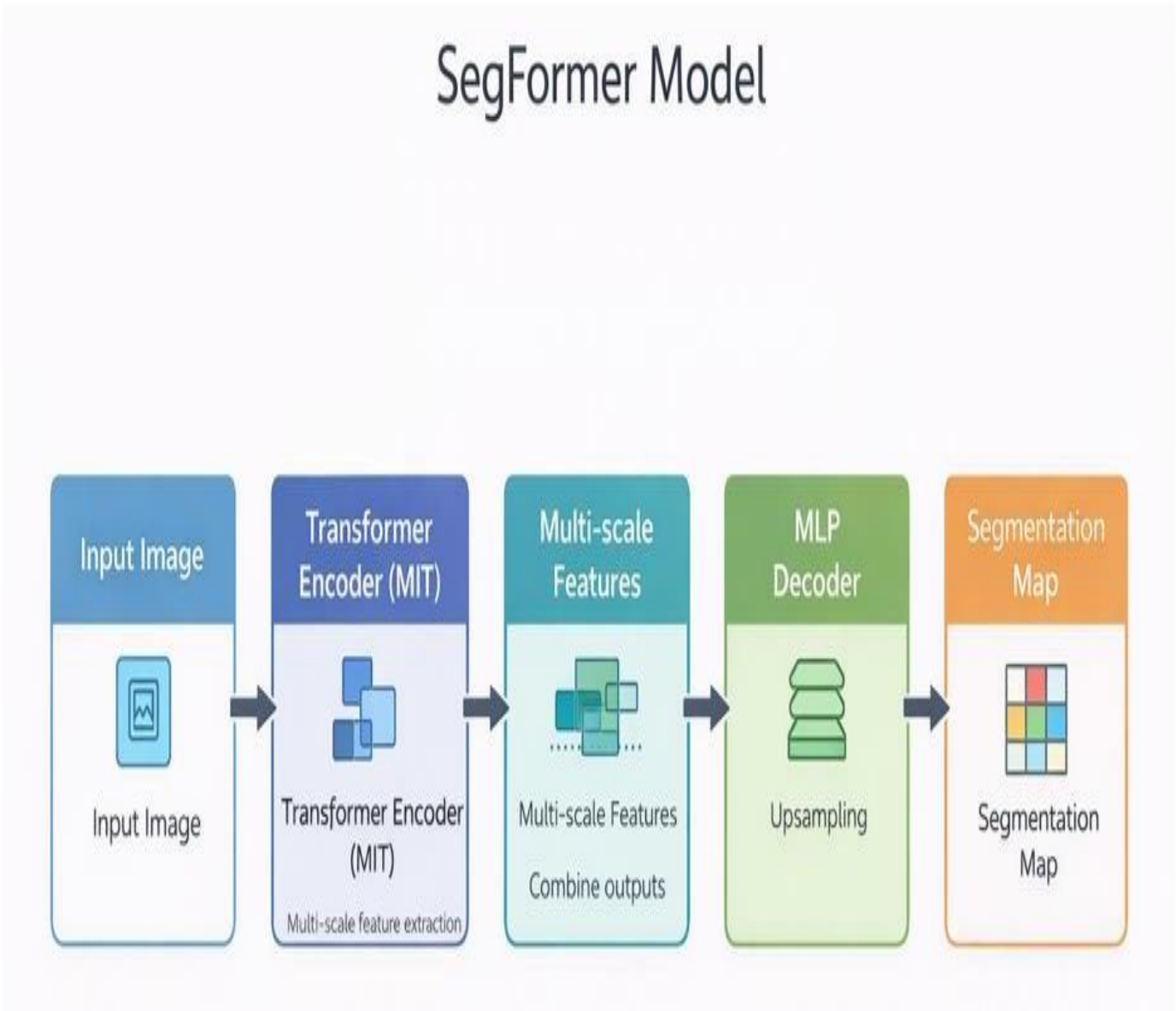


Figure 3.3 SegFormer Model

1. Hierarchical MiT Encoder (Mix Transformer)

- Works in **four stages**, progressively reducing resolution while increasing semantic understanding.
- Stage 1: High-resolution local details (edges of cracks).
- Stage 2 & 3: Mid-level features (patterns, textures).
- Stage 4: Global context (overall wall structure).
- Uses **overlapping patch merging** instead of positional encoding, making it robust to different camera resolutions and input sizes.

2. All-MLP Decoder

- Extremely lightweight compared to CNN-based decoders.
- Takes multi-scale features from all encoder stages.
- Uses **MLPs (linear layers)** to unify dimensions, then upsamples and concatenates them.
- Produces a **pixel-level segmentation mask** with minimal computation.

Why SegFormer is Special

- **Efficiency:** Runs on low-power devices like Raspberry Pi without needing GPUs or cloud servers.
- **Resolution Independence:** Works even if the input image size changes, unlike traditional transformers.
- **Accuracy for Thin Structures:** Preserves fine, branching crack details that bounding-box methods miss.
- **Scalability:** Can be adapted to different environments and camera setups.

Role in Project

In your system, SegFormer is the **vision branch**:

- It captures **precise crack geometry** (length, width, branching).
- Provides a **topology-based severity score** rather than crude bounding boxes.
- Its output is later fused with **acoustic analysis** to form a unified risk score.

3.4 . Data Preprocessing

1. Image Data (Vision Branch)

- Image Acquisition: Capture raw images of walls/cracks using the Pi Camera.
- Resizing & Normalization: Standardize image dimensions and pixel values for SegFormer input.
- Noise Reduction: Apply filters to remove background noise or lighting artifacts.
- Annotation & Labeling: Prepare datasets with pixel-level labels (crack vs. wall vs. background).
- Augmentation: Use techniques like rotation, flipping, and scaling to increase dataset diversity and improve model robustness.

2. Acoustic Data (FFT Branch)

- Signal Recording: Collect tap-test responses using the microphone.
- Noise Filtering: Remove environmental noise and normalize amplitude.
- Windowing: Apply signal window functions (e.g., Hamming window) to prepare for FFT.
- FFT Transformation: Convert time-domain signals into frequency-domain features.
- Feature Extraction: Identify resonance peaks that indicate hollowness or delamination.

3. IMU Data (Alignment Control)

- Orientation Capture: Collect accelerometer and gyroscope readings.
- Thresholding: Verify device is orthogonal to the wall before data collection.
- Filtering: Smooth IMU signals to reduce jitter and ensure stable alignment.

4. Fusion Preprocessing

- Normalization: Scale vision and acoustic scores to a common range.
- Timestamping & Geo-tagging: Attach metadata for temporal tracking and location-based analysis.
- Database Storage: Store preprocessed outputs in the onboard temporal database for prognostic tracking.

3.5 . Modules of the Project

1. IMU-Based Orthographic Lock Module

- Ensures the device is perfectly aligned with the wall before scanning.
- Eliminates perspective distortion caused by human operator error.

2. Vision Analysis Module (SegFormer)

- Captures images using the Pi Camera.
- Performs pixel-level crack segmentation and topology extraction.
- Generates a vision-based severity score based on crack geometry.

3. Acoustic Analysis Module (FFT Processing)

- Records tap-test responses through a microphone.
- Applies FFT to detect subsurface defects such as hollowness or delamination.
- Produces an acoustic resonance score indicating internal structural integrity.

4. Multimodal Sensor Fusion Module

- Combines vision and acoustic scores using a deterministic fusion algorithm.
- Outputs a unified, explainable risk classification score (0–10).

5. Prognostic Tracking Module

- Stores inspection results in an onboard temporal database.
- Tracks degradation velocity and growth trends over time.
- Enables predictive maintenance rather than reactive repairs.

6. Geo-Tagging & Reporting Module

- Attaches timestamps and location data to inspection results.
- Generates a comprehensive structural health report.
- Displays results locally on the edge device without cloud dependency.

CHAPTER 4

CONCLUSION AND FUTURE ENHANCEMENT AND REFERENCES

Conclusion

The project successfully demonstrates the design of an IMU-Locked Multimodal Edge Device for structural health prognostics, integrating vision-based crack segmentation and acoustic subsurface analysis into a unified, offline diagnostic tool. By eliminating cloud dependency, ensuring orthographic alignment, and introducing temporal prognostic tracking, the system addresses critical gaps in existing methods such as subjectivity, perspective distortion, and lack of predictive insights. The device provides a low-cost, scalable, and reliable solution for infrastructure monitoring, particularly in rural and resource-constrained environments, thereby contributing to safer communities and proactive maintenance strategies.

Future Enhancements

- **Enhanced Model Deployment:** Upgrade to more advanced lightweight segmentation models for improved accuracy.
- **Expanded Acoustic Analysis:** Incorporate machine learning classifiers to distinguish between different defect types beyond hollowness.
- **IoT Integration:** Enable optional cloud connectivity for centralized monitoring in urban deployments.
- **Mobile App Interface:** Develop a companion app for real-time visualization, reporting, and alerts.
- **Battery Optimization:** Improve power efficiency for longer field usage in remote areas.
- **Multi-Surface Adaptability:** Extend functionality to diverse materials (concrete, steel, wood).
- **Edge AI Acceleration:** Explore hardware accelerators (e.g., Coral TPU, NVIDIA Jetson) for faster inference.

Field Visit and Preventive Measures Observed



Figure 4.1 Field Visit and Preventive Measures Observed

As part of the project study, a visit was conducted to a local construction site to understand the practical causes of structural cracks and the preventive measures followed by experienced engineers during construction. The engineers explained that crack prevention starts from the initial construction stage itself through proper planning, quality materials, and correct construction practices. They emphasized maintaining the correct cement–sand–water ratio, using high-quality construction materials, proper steel reinforcement placement, and adequate curing of concrete to reduce shrinkage cracks and improve structural strength. They also highlighted the importance of soil testing, waterproofing, load management, and providing expansion joints to prevent cracks caused by settlement, moisture, and temperature variations. Although these precautions significantly reduce the chances of crack formation, the engineers noted that cracks can still occur over time due to aging, environmental stress, overloading, and poor maintenance. This field study helped us understand the importance of continuous structural monitoring and supported the need for our proposed crack detection and risk assessment system.

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